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Orienting perforating gun system, and method of orienting shots in a perforating gun assembly

Abstract

A perforating gun system. The perforating gun system comprises a gun barrel housing, and a pair of tandem subs threadedly connected to the gun barrel housing at opposing ends. The perforating gun system also includes a rail. The rail defines an elongated frame having a plurality of receptacles, wherein each receptacle is configured to receive a shaped charge. A ballast is connected to each of opposing ends of the rail. Each ballast supports a bearing member which interfaces with a central bore of a respective tandem sub. In this manner, relative rotation is provided between the rail and the first and second tandem subs. A method of orienting shots in a perforating gun assembly is also provided.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3181608	12/1964	Palmer	N/A	N/A
3605918	12/1970	Bennett	N/A	N/A
4194577	12/1979	Vann	N/A	N/A
4269278	12/1980	Vann	N/A	N/A
5571986	12/1995	Snider et al.	N/A	N/A
5869968	12/1998	Brooks et al.	N/A	N/A
5911277	12/1998	Hromas et al.	N/A	N/A
6837310	12/2004	Martin	N/A	N/A
7213655	12/2006	Parrott	N/A	N/A
7886842	12/2010	Howard et al.	N/A	N/A
8028751	12/2010	Pinto et al.	N/A	N/A
8186259	12/2011	Burleson et al.	N/A	N/A
8684083	12/2013	Torres et al.	N/A	N/A
9115572	12/2014	Hardesty et al.	N/A	N/A
9399897	12/2015	Stokes	N/A	N/A
9500071	12/2015	Morgan-Smith et al.	N/A	N/A
9523266	12/2015	Arena	N/A	N/A
9581422	12/2016	Preiss et al.	N/A	N/A
9598942	12/2016	Wells et al.	N/A	N/A
9644460	12/2016	Bell et al.	N/A	N/A
9689223	12/2016	Schacherer et al.	N/A	N/A
9784549	12/2016	Eitschberger	N/A	N/A
9803455	12/2016	Yang et al.	N/A	N/A
9845666	12/2016	Hardesty et al.	N/A	N/A
10036236	12/2017	Sullivan et al.	N/A	N/A
10062476	12/2017	Varkey et al.	N/A	N/A
10113401	12/2017	Al-Gouhi	N/A	N/A
10233743	12/2018	Morgan-Smith et al.	N/A	N/A
10273760	12/2018	Cook et al.	N/A	N/A
10273788	12/2018	Bradley et al.	N/A	N/A

10316634	12/2018	Graham et al.	N/A	N/A
10337310	12/2018	Bell et al.	N/A	N/A
10352136	12/2018	Goyeneche	N/A	N/A
10358897	12/2018	Bowen et al.	N/A	N/A
10392888	12/2018	Robertson et al.	N/A	N/A
10444392	12/2018	Milne et al.	N/A	N/A
10458213	12/2018	Eitschberger et al.	N/A	N/A
10465488	12/2018	Collins et al.	N/A	N/A
10472937	12/2018	Walters et al.	N/A	N/A
10526876	12/2019	Hess et al.	N/A	N/A
10584950	12/2019	Saltarelli et al.	N/A	N/A
10590706	12/2019	Hess et al.	N/A	N/A
10597979	12/2019	Eitschberger et al.	N/A	N/A
10683740	12/2019	Randall et al.	N/A	N/A
10689955	12/2019	Mauldin et al.	N/A	N/A
10731444	12/2019	Wells et al.	N/A	N/A
10844696	12/2019	Eitschberger et al.	N/A	N/A
10844697	12/2019	Preiss et al.	N/A	N/A
D904475	12/2019	Preiss et al.	N/A	N/A
D908754	12/2020	Eitschberger et al.	N/A	N/A
10954769	12/2020	Randall et al.	N/A	N/A
10975671	12/2020	Bradley et al.	N/A	N/A
D920402	12/2020	Eitschberger et al.	N/A	N/A
D921858	12/2020	Eitschberger et al.	N/A	N/A
D922541	12/2020	Mulhern et al.	N/A	N/A
11053782	12/2020	Loehken et al.	N/A	N/A
11078762	12/2020	Mauldin et al.	N/A	N/A
11125056	12/2020	Parks et al.	N/A	N/A
D935574	12/2020	Eitschberger et al.	N/A	N/A
11225848	12/2021	Eitschberger	N/A	N/A
11248452	12/2021	Sullivan et al.	N/A	N/A
11255650	12/2021	Sullivan et al.	N/A	N/A
11293737	12/2021	Sullivan et al.	N/A	N/A
11299967	12/2021	Bradley et al.	N/A	N/A
D951207	12/2021	Donauer	N/A	N/A
11402190	12/2021	Sullivan et al.	N/A	N/A
11428081	12/2021	Bradley et al.	N/A	N/A
11480038	12/2021	Eitschberger	N/A	N/A
D968474	12/2021	Volberg	N/A	N/A
11536118	12/2021	Sullivan et al.	N/A	N/A
11624266	12/2022	Mauldin et al.	N/A	N/A
11629579	12/2022	Davis et al.	N/A	N/A
11668166	12/2022	Lopez de Cardenas et al.	N/A	N/A
11674371	12/2022	Bradley et al.	N/A	N/A
D994736	12/2022	Sullivan	N/A	N/A
11713661	12/2022	Oehring et al.	N/A	N/A
11725506	12/2022	Shahkarami et al.	N/A	N/A
11753917	12/2022	Shetty et al.	N/A	N/A
11795791	12/2022	Eitschberger et al.	N/A	N/A

11795792	12/2022	Archibald et al.	N/A	N/A
11834940	12/2022	Kohli et al.	N/A	N/A
11906278	12/2023	Sullivan	N/A	N/A
11913767	12/2023	Sullivan et al.	N/A	N/A
2003/0047358	12/2002	Bonkowski	N/A	N/A
2003/0098158	12/2002	George et al.	N/A	N/A
2004/0144539	12/2003	Smith et al.	N/A	N/A
2011/0132607	12/2010	Lahitette et al.	N/A	N/A
2012/0061098	12/2011	Hall	N/A	N/A
2014/0020896	12/2013	Al-Gouhi	N/A	N/A
2014/0131035	12/2013	Entchev et al.	N/A	N/A
2014/0137723	12/2013	Umphries et al.	N/A	N/A
2014/0158432	12/2013	Simpson	N/A	N/A
2015/0000509	12/2014	Current et al.	N/A	N/A
2015/0330192	12/2014	Rogman et al.	N/A	N/A
2016/0024858	12/2015	Hanton et al.	N/A	N/A
2016/0061572	12/2015	Eitschberger et al.	N/A	N/A
2016/0084048	12/2015	Harrigan et al.	N/A	N/A
2016/0168961	12/2015	Parks et al.	N/A	N/A
2016/0237752	12/2015	Jones	N/A	N/A
2016/0333675	12/2015	Wells et al.	N/A	N/A
2017/0051586	12/2016	Wells et al.	N/A	N/A
2017/0074078	12/2016	Eitschberger	N/A	N/A
2017/0211363	12/2016	Bradley et al.	N/A	N/A
2017/0241209	12/2016	Benes	N/A	N/A
2018/0016846	12/2017	Peter et al.	N/A	N/A
2018/0058167	12/2017	Finol et al.	N/A	N/A
2018/0112465	12/2017	Ritchie et al.	N/A	N/A
2018/0112524	12/2017	Huang et al.	N/A	N/A
2018/0119529	12/2017	Goyeneche	N/A	N/A
2018/0135389	12/2017	Sullivan	N/A	N/A
2018/0347324	12/2017	Langford et al.	N/A	N/A
2019/0048693	12/2018	Henke et al.	N/A	N/A
2019/0136673	12/2018	Sullivan et al.	N/A	N/A
2019/0145216	12/2018	Shampine	N/A	N/A
2019/0153829	12/2018	Roessler	N/A	N/A
2019/0153841	12/2018	Randall et al.	N/A	N/A
2019/0211655	12/2018	Bradley et al.	N/A	N/A
2019/0242199	12/2018	Al-Qasim et al.	N/A	N/A
2019/0257181	12/2018	Langford et al.	N/A	N/A
2019/0264548	12/2018	Zhao et al.	N/A	N/A
2019/0277093	12/2018	Morin	N/A	N/A
2020/0024935	12/2019	Eitschberger et al.	N/A	N/A
2020/0063537	12/2019	Langford et al.	N/A	N/A
2020/0157924	12/2019	Melhus et al.	N/A	N/A
2020/0284104	12/2019	Holmberg et al.	N/A	N/A
2020/0392821	12/2019	Eitschberger et al.	N/A	N/A
2020/0399995	12/2019	Preiss et al.	N/A	N/A
2021/0131205	12/2020	Tran	N/A	N/A
2021/0270104	12/2020	Løvoll et al.	N/A	N/A

2021/0301599	12/2020	Mulhern et al.	N/A	N/A
2022/0003085	12/2021	Loehken	N/A	N/A
2022/0018638	12/2021	Bergmann et al.	N/A	N/A
2022/0049588	12/2021	Sampson	N/A	N/A
2022/0195824	12/2021	Scharf et al.	N/A	N/A
2022/0205344	12/2021	Hoelscher et al.	N/A	N/A
2022/0243567	12/2021	Eitschberger et al.	N/A	N/A
2022/0268135	12/2021	Eitschberger et al.	N/A	N/A
2022/0282578	12/2021	Eitschberger	N/A	N/A
2022/0307330	12/2021	Eitschberger et al.	N/A	N/A
2022/0316283	12/2021	Eitschberger et al.	N/A	N/A
2022/0356793	12/2021	Zitting	N/A	N/A
2022/0403718	12/2021	Eitschberger et al.	N/A	N/A
2022/0412195	12/2021	Vriend et al.	N/A	N/A
2023/0003106	12/2022	Conzemius	N/A	N/A
2023/0193711	12/2022	Eitschberger et al.	N/A	N/A
2023/0203896	12/2022	Yard	N/A	N/A
2023/0203923	12/2022	Eitschberger	N/A	N/A
2023/0220731	12/2022	Moyes	N/A	N/A
2023/0265748	12/2022	Bradley et al.	N/A	N/A
2023/0266109	12/2022	Teowee et al.	N/A	N/A
2023/0296364	12/2022	Teowee et al.	N/A	N/A
2023/0304384	12/2022	Rajaram et al.	N/A	N/A
2023/0323739	12/2022	Eitschberger	N/A	N/A
2023/0332641	12/2022	Moyes	N/A	N/A
2023/0349238	12/2022	Martin	N/A	N/A
2023/0349272	12/2022	Blois et al.	N/A	N/A
2023/0349677	12/2022	Scharf et al.	N/A	N/A
2023/0358105	12/2022	Oettli	N/A	N/A
2023/0366298	12/2022	Sullivan	N/A	N/A
2023/0366299	12/2022	Badii et al.	N/A	N/A
2023/0374892	12/2022	Dyess et al.	N/A	N/A
2023/0383625	12/2022	Eitschberger et al.	N/A	N/A
2023/0392592	12/2022	Shampine et al.	N/A	N/A
2024/0035354	12/2023	McWilliam	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2698970	12/2004	CN	N/A

OTHER PUBLICATIONS

Machine translation of the Abstract of Foreign Reference: CN 2698970 Y, Published: May 11, 2005; Baoji Oilfield Machinery Co Ltd; 1 page. cited by applicant

U.S. Appl. No. 29/920,858, filed Dec. 13, 2023; Inventors: Shelby L. Sullivan and Joshua M. Scott; 17 pages. cited by applicant

U.S. Appl. No. 29/920,815, filed Dec. 13, 2023; Inventors: Shelby L. Sullivan and Joshua M. Scott; 7 pages. cited by applicant

An identification of earlier applications pursuant to 37 CFR 1.98 (d) (1) is attached listing: U.S. Appl. No. 29/920,815; U.S. Appl. No. 29/920,815; and U.S. Pat. No. 11,906,278; 1 page. cited by applicant

Examination Report issued in related CA Application No. 226994; Issued: Dec. 5, 2024; 6 pages. cited by applicant
Website: Youtube.com—Hind Bearings; Publication date: Dec. 21, 2020; Link: https://youtube.com/shorts/IJQViiJk_h0?si=ShTbizwPxdF1VZ16; 1 page. cited by applicant
Website: Youtube.com—KG Bearing India; Publication date: Dec. 14, 2022; Link: <https://youtube.com/shorts/H2fThg6jukk?si=1OKSaAZ7vYYJSxbC>; 1 page. cited by applicant
Website: Youtube.com—VXB Ball Bearings; Publication date: Mar. 20, 2023; Link: https://youtube.com/shorts/FBbQOVYtNvQ?si=soMtRfWlzaa1_PEU; 1 page. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of U.S. Ser. No. 63/497,305. That application was filed on Apr. 20, 2023 and was entitled “Orienting Perforating Gun System, and Method of Orienting Shots in a Perforating Gun Assembly.” (2) The present application also claims the benefit of U.S. Ser. No. 63/511,903. That application was filed on Jul. 5, 2023 and was entitled “Orienting Perforating Gun System, and Method of Orienting Shots in a Perforating Gun Assembly.” (3) The present application is filed as a continuation-in-part of U.S. Ser. No. 29/920,858 filed Dec. 13, 2023. That application is titled “Tandem Sub For A Roller Bearing.” (4) The present application is filed as a continuation-in-part of U.S. Ser. No. 29/920,815 filed Dec. 13, 2023. That application is titled “Roller Bearing For A Tandem Sub.” (5) Each of these provisional filings is incorporated herein in its entirety by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not applicable.

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

(2) Not applicable.

BACKGROUND OF THE INVENTION

(3) This section is intended to introduce various aspects of the art, which may be associated with exemplary embodiments of the present disclosure. This discussion is believed to assist in providing a framework to facilitate a better understanding of particular aspects of the present invention. Accordingly, it should be understood that this section should be read in this light, and not necessarily as admissions of prior art.

FIELD OF THE INVENTION

(4) The present disclosure relates to the field of hydrocarbon recovery operations. More specifically, the disclosure relates to the completion of a well for the production of oil and gas. More specifically still, the inventions herein relate to a perforating gun assembly wherein the shots along the perforating guns may be selectively oriented.

TECHNOLOGY IN THE FIELD OF THE INVENTION

(5) In the drilling of an oil and gas well, a near-vertical wellbore is formed through the earth using a drill bit urged downwardly at a lower end of a drill string. After drilling to a predetermined depth, the drill string and bit are removed and the wellbore is lined with a string of casing. An annular area is thus formed between the string of casing and the formation penetrated by the wellbore.

(6) A cementing operation is conducted in order to fill or “squeeze” the annular area with cement along part or all of the length of the wellbore. The combination of cement and casing strengthens

the wellbore and facilitates zonal isolation, and subsequent completion, of hydrocarbon-producing pay-zones behind the casing.

(7) In connection with the completion of the wellbore, several strings of casing having progressively smaller outer diameters will be cemented into the wellbore. These will include a string of surface casing, one or more strings of intermediate casing, and finally a string of production casing. The process of drilling and then cementing progressively smaller strings of casing is repeated until the well has reached total depth (“TD”). In some instances, the final string of casing is a liner, that is, a string of casing that is not tied back to the surface.

(8) Within the last two decades, advances in drilling technology have enabled oil and gas operators to “kick-off” and steer wellbore trajectories from a vertical orientation to a near-horizontal orientation. The horizontal “leg” of each of these wellbores now often exceeds a length of one mile, and sometimes two or even three miles. This significantly multiplies the wellbore exposure to a target hydrocarbon-bearing formation. The horizontal leg will typically include the production casing.

(9) FIG. 1 is a side, cross-sectional view of a wellbore **100**, in one embodiment. The wellbore **100** has been completed horizontally, that is, it is completed with a horizontal leg **156**. The wellbore **100** defines a bore **10** that has been drilled from an earth surface **105** into a subsurface **110**. The wellbore **100** is formed using any known drilling mechanism, but preferably using a land-based rig or an offshore drilling rig on a platform.

(10) The wellbore **100** is completed with a first string of casing **120**, sometimes referred to as surface casing. The wellbore **100** is further completed with a second string of casing **130**, typically referred to as an intermediate casing. In deeper wells, that is wells completed 7,500 feet or more below the earth surface **105**, at least two intermediate strings of casing will be used. In FIG. 1, a second intermediate string of casing is shown at **140**. Together, casings **130**, **140** comprise intermediate casing strings.

(11) The wellbore **100** is finally completed with a string of production casing **150**. In the view of FIG. 1, the production casing **150** extends from the surface **105** down to a subsurface formation, or “pay-zone” **115**. As noted, the wellbore **100** is completed horizontally, meaning that a horizontal portion (indicated by leg **156**) is provided. The horizontal portion **156** includes a heel **153** and a toe **154**. In this instance, the toe **154** defines the end (or “TD”) of the wellbore **100**.

(12) It is observed that the annular region around the surface casing **120** is filled with cement **125**. The cement (or cement matrix) **125** serves to isolate the wellbore from fresh water zones and potentially porous formations around the production casing **120**.

(13) The annular regions around the intermediate casing strings **130**, **140** are also filled with cement **135**, **145**. Similarly, the annular region around the production casing **150** is filled with cement **155**. However, the cement **135**, **145**, **155** is optionally only placed behind the respective casing strings **130**, **140**, **150** up to the lowest joints of the immediately surrounding casing strings. Thus, a non-cemented annular region **132** is typically preserved above the cement matrix **135**; a non-cemented annular region **142** is typically preserved above the cement matrix **145**; and a non-cemented annular region **152** is frequently preserved above the cement matrix **155**.

(14) In order to facilitate the recovery of hydrocarbons, particularly in low-permeability formations **115**, the production casing **150** along the horizontal portion **156** undergoes a process of perforating and fracturing (or in some cases perforating and acidizing). Due to the very long lengths of new horizontal wells, the perforating and formation treatment process is carried out in multiple stages.

(15) In one method, a perforating gun assembly **200** is pumped down the wellbore **100** towards the toe **154** at the end of a wireline **240**. The perforating gun assembly **200** will include a series of perforating guns (shown at **210** in FIG. 2), with each perforating gun **210** having sets of charges ready for detonation. The charges associated with one of the perforating guns **210** are detonated, and perforations are “shot” into the casing **150**. Those of ordinary skill in the art will understand that the perforating guns **210** have explosive charges, typically shaped charges, which are ignited to

create holes in the casing **150** (and, if present, the surrounding cement) and pass at least a few inches, and possibly several feet, into the formation **115**. The perforations (not shown) create fluid communication with the surrounding formation **115** such that hydrocarbon fluids can flow out of the formation **115**, into the casing **150**, and up to the surface **105**.

(16) After perforating, the operator will fracture the formation **115** through the perforations. This is done by pumping treatment fluids into the formation **115** at a pressure above a formation parting pressure. After the fracturing operation is complete, the wireline **240** will be raised and the perforating gun assembly **200** will be positioned at a new location (or “depth”) along the horizontal portion **156**. A plug (such as plug **112** of FIG. **1**) is set below the perforating gun assembly **200** and new shots are fired in order to create a new set of perforations. Thereafter, treatment fluid is again pumped into the wellbore **100** and into the formation **115** at a pressure above the formation parting pressure. In this way, a second set of fractures is formed away from the wellbore **156**.

(17) The process of setting a plug, perforating the casing, and fracturing the formation is repeated in multiple stages until the wellbore **100** has been completed, that is, the wellbore **100** is ready for production. The shots create clusters of perforations to create fracture complexity and to enhance fluid communication with the formation **115**. The result is that multiple plugs **112** are set along the horizontal wellbore **156** during the completion process.

(18) In order to provide perforations for the multiple stages without having to pull the perforating gun after each detonation, the perforating gun assembly **200** employs multiple guns **210** in series. FIG. **2** is a side view of an illustrative perforating gun assembly **200**, or at least a portion of an assembly. The perforating gun assembly **200** comprises a string of individual perforating guns **210**.

(19) Each perforating gun **210** represents various components. These typically include a “gun barrel” **212** which serves as an outer tubular housing. An uppermost gun barrel housing **212** is supported by an electric wire (or “e-line”) **240** that extends from the surface **105** and that delivers electrical energy down to the perforating gun assembly **200**. Each perforating gun **210** also includes an explosive initiator, or “detonator” (shown in phantom at **229**).

(20) In addition, each perforating gun **210** comprises a detonating cord (shown at **545** in FIG. **5A**). The detonating cord **545** contains an explosive compound that is ignited by the detonator **229**. Thus, the detonator **229** receives electrical energy which in turn initiates the detonator cord **545**. The detonator cord **545** propagates an explosion down its length to a series of charges (typically referred to as shaped charges). In some cases, the shaped charges are held in an inner tube, referred to as a charge carrier tube, for security. The charges are discharged through openings **215** in the selected gun barrel **212**. An illustrative charge carrier tube is shown at **500** of FIGS. **5** and **6A** of co-owned U.S. Pat. No. **11,293,737**, which is incorporated herein in its entirety by reference.

(21) The perforating gun assembly **200** may include short centralizer subs **220**. In addition, tandem subs **225** may be used to connect the gun barrel housings **212** end-to-end. Each tandem sub **225** comprises a metal threaded connector placed between the gun barrels **210**. Typically, the gun barrel housings **212** will have female-by-female threaded ends (such as end **412** in FIG. **5B**) while the tandem subs **225** have opposing male threaded ends.

(22) Further, an insulated connection member **230** connects the e-line **240** to the uppermost gun barrel **210**. The e-line **240** extends upward to a control interface (not shown) located at the surface **105**.

(23) The perforating gun assembly **200** and its long string of gun barrels (the housings **212** of the perforating guns **210**) are carefully assembled at the surface **105**, and then lowered into the wellbore **100** at the end of the e-line **240**. After the casing **150** has been perforated and at least one plug **112** has been set, the setting tool **160** and the perforating gun assembly **200** are taken out of the wellbore **100**, and a ball (not shown) is dropped into the wellbore **100** to close the plug **112**. When the plug **112** is closed, a frac slurry (e.g., a mixture of water, sand, and surfactant) is pumped by a pumping system down the wellbore **100** for fracturing purposes. For a formation fracturing operation, the pumping system will create downhole pressure that is above the formation parting

pressure.

(24) It is observed that when the operator sends the detonation signals downhole, the directions to which the shots are fired into the formation cannot be controlled; instead, only the depth at which the shots are fired is controllable. In this respect, when the operator releases a perforating gun string into the wellbore **100**, the perforating gun assembly **200** may (and likely will) rotate as it gravitationally falls into the wellbore **100**. As the assembly **200** is pumped across the heel **153** and through the horizontal portion **156**, additional rotational movement may occur. However, the operator may desire that shots be fired not only at selected depths, but also at a selected orientation about the central axis of a generally horizontal wellbore. Specifically, operators may prefer that the perforations be formed vertically upward (a 0° orientation) or downward (a 180° orientation) perpendicular from the central axis of the wellbore. Alternatively, operators may prefer that the perforations be formed laterally outward (90° or 270° orientation) from the central axis of the wellbore. This urges the fractures to propagate outwardly from the wellbore **100** in the direction of the perforations.

(25) A need exists for an improved perforating gun assembly with orienting capability allowing perforations in any desired orientation from 0° to 360° about the central axis of the wellbore. Further, a need exists for an improved method of aligning charges along a perforating gun assembly for use in a wellbore. Still further, a need exists for a method of avoiding frac hits by shooting aligned perforations in a specific desired direction which may be away from a parent wellbore.

BRIEF SUMMARY OF THE INVENTION

(26) A novel perforating gun assembly is first provided herein. The perforating gun assembly is arranged to enable the operator the ability to select the orientation in which shots may be fired downhole. Shots may be fired in connection with the completion of a well used for the production of hydrocarbon fluids. Alternatively, shots may be fired in connection with the completion of a well used for the production of steam, that is, for a geothermal well.

(27) In one aspect, the perforating gun assembly first includes a gun barrel housing. The gun barrel housing has a first end, a second end, and a bore extending between the first and second ends. Thus, the gun barrel housing defines a tubular body.

(28) The perforating gun assembly also includes a pair of tandem subs. Specifically, a tandem sub is placed at each of the opposing ends of the gun barrel housing. Each tandem sub has a pair of male threaded ends and a bore extending therein. The tandem subs are threadedly connected to the respective ends of adjacent gun barrel housings.

(29) The perforating gun assembly also includes a rail system. The rail system resides within a gun barrel housing. The rail system comprises: a rail defining an elongated frame; a series of receptacles placed along the frame; a charge (such as a shaped charge) residing within each of the series of receptacles; a first ballast secured to a first end of the rail; and a second ballast secured to a second end of the rail.

(30) Uniquely, the perforating gun system provides for a bearing connection. In this respect, a first bearing member resides at a distal end of the first ballast while a second bearing member resides at a distal end of the second ballast. Each ballast member engages and rotates within the bore of a respective tandem sub. Thus, when the perforating gun assembly is pumped into the horizontal leg of a wellbore, the ballasts will rotate into a downward position, placing the shaped charges into desired orientations.

(31) Beneficially, the rail system may be used to support the charges in lieu of using a so-called “charger carrier tube.” The charges may be inserted into the respective receptacles along the rail from either a forward side or a rearward side of the frame. Additionally, the ballasts may be connected to the rail at varying angles relative to the frame. In this way, shots may be fired within the horizontal section of a wellbore in a pre-selected orientation relative to the central axis of the wellbore (or the central axis of the gun barrel Housing).

(32) A perforating gun system is also provided herein. The perforating gun system also includes a

gun barrel housing. The gun barrel housing has a first end, a second end opposite the first end, and a bore extending from the first end to the second end of the gun barrel housing. Thus, the gun barrel housing defines a tubular body.

(33) The perforating gun system also comprises a rail. The rail defines an elongated frame having a plurality of receptacles. Each receptacle is configured to receive a charge (such as a shaped charge). Each of the shaped charges may be received within a respective receptacle from either side of the frame. In this way, the charges may face opposite directions relative to each other within the wellbore, or they may all face in the same direction.

(34) In one aspect, the rail comprises one to five receptacles, meaning one to five charges are received. Preferably, the rail is fabricated from aluminum, an aluminum alloy, or a rigid polymeric material. In one aspect, two rails can be connected end-to-end, effectively forming one longer rail with additional receptacles.

(35) A first flange resides at a first end of the rail. Similarly, a second flange resides at a second end of the rail. Each flange includes a plurality of through-openings spaced equi-radially around a perimeter.

(36) The perforating gun system also includes a pair of tandem subs. A first tandem sub is threadedly connected to the first end of the gun barrel housing, while a second tandem sub is threadedly connected to the second end of the gun barrel housing. Each of the first and second tandem subs comprises a first end, a second end opposite the first end, and a bore extending from the first end to the second end.

(37) Preferably, each of the first tandem sub and the second tandem sub comprises male threaded ends. At the same time, the first and second opposing ends of the gun barrel housing each comprises female threads, forming a female-by-female tubular body. In one aspect, each tandem sub houses an addressable switch.

(38) Additionally, the perforating gun system includes a pair of ballasts. These represent a first ballast and a second ballast. Each of the first and second ballasts comprises a weighted body. The first ballast comprises a proximal end connected to the first flange, and a distal end bearingly abutting the first tandem sub. At the same time, the second ballast comprises a proximal end connected to the second flange, and a distal end bearingly abutting the second tandem sub. The weighted bodies reside between the proximal and distal ends of each of the ballasts.

(39) The perforating gun system takes advantage of a pair of bearing connections. In this respect, the system offers a first bearing member supported by the first ballast at the distal end of the first ballast. The system then offers a second bearing member supported by the second ballast at its distal end. The first bearing member interfaces with and rotates within the bore of the first tandem sub, while the second bearing member interfaces with and rotates within the bore of the second tandem sub. The result of these interfaces is that the perforating gun system allows for relative rotation between the ballasts and the first and second tandem subs. This, in turn, allows for relative rotation of the rail and supported shaped charges within the gun barrel housing.

(40) The shaped charges may be placed within respective receptacles along the frame to fire at orientations of any of 0 degrees, 90 degrees, 180 degrees, or 270 degrees within the horizontal wellbore, depending on the direction in which the charges are inserted into the respective receptacles, depending on the angle at which the ballasts are secured to the flanges of the rail. Intermediate angles (such as at 45 degree angles) may be selected at the operator's preference.

(41) In one aspect, a first electrically conductive contact pin resides within the bore of the first tandem sub. Similarly, a second electrically conductive contact pin resides within the bore of the second tandem sub. A detonator resides within the gun barrel housing adjacent the rail and is in electrical communication with the contact pins.

(42) A method of orienting shots in a perforating gun assembly is also provided herein. In one aspect, the method first comprises providing a perforating gun assembly. The perforating gun assembly may be arranged in accordance with any of the embodiments described above. In this

respect, the perforating gun assembly includes a gun barrel housing and a pair of tandem subs. Specifically, a tandem sub is placed at each of the opposing ends of the gun barrel housing. (43) The method also includes providing a rail. The rail defines an elongated frame having a series of receptacles along its length. The receptacles are configured to receive respective charges. Preferably, the receptacles each have a circular profile and are equi-distantly spaced along the frame.

(44) The method further comprises placing a charge within each of the respective receptacles along the frame. Preferably, the rail includes at least three receptacles. Beneficially, the charges may be received within a respective receptacle from either side of the frame.

(45) The method additionally comprises threadedly connecting the first tandem sub to the first end of the gun barrel housing. The method then includes threadedly connecting the second tandem sub to the second end of the gun barrel housing.

(46) The perforating gun system includes a pair of ballasts. These represent a first ballast and a second ballast. Each of the first and second ballasts comprises a weighted body. The first ballast comprises a distal end abutting the first tandem sub. At the same time, the second ballast comprises a distal end abutting the second tandem sub.

(47) The method also comprises providing a first bearing connection between the first ballast and the first tandem sub. Additionally, the method comprises providing a second bearing connection between the second ballast and the second tandem sub. In this way, the first ballast, the second ballast, and the connected rail may rotate together relative to the first and second tandem subs and the connected gun barrel housing.

(48) In one aspect, each of the first and second tandem subs comprises a first end and a second end opposite the first end. Each of the first and second tandem subs further comprises a bore extending from the first end to the second end of the respective tandem sub. The first bearing member interfaces with and rotates within the bore of the first tandem sub, while the second bearing member interfaces with and rotates within the bore of the second tandem sub. This allows for the relative rotation between the rail and the first and second tandem subs.

(49) In one embodiment, the method also includes placing each of the shaped charges into respective charge jackets. Each charge jacket is fabricated from a compliant material, allowing the charges to be removably snapped into place. The rail further comprises slots associated with each of the receptacles. Each of the charge jackets comprises side rails configured to be received within the slots of a respective receptacle along the frame. In this way, the shaped charges are held securely in position along the frame.

(50) The method may also include selecting a direction for each of the shaped charges within its respective receptacle along the frame. The method then includes connecting the proximal end of the first ballast to the first end of the rail, and connecting the proximal end of the second ballast to the second end of the rail.

(51) In one embodiment, the rail comprises a first end defining a first flange, and a second end defining a second flange. The first flange comprises at least four equi-radially spaced through-openings. Similarly, the second flange also comprises at least four equi-radially spaced through-openings. The proximal end of the first ballast comprises a pair of through-openings that may be aligned with a selected pair of the at least four equi-radially spaced through-openings of the first flange. At the same time, the proximal end of the second ballast comprises a pair of through-openings that may be aligned with a selected pair of the at least four equi-radially spaced through-openings of the second flange. In this arrangement, connecting the proximal end of the first ballast to the first end of the rail, and connecting the proximal end of the second ballast to the second end of the rail, comprises: (i) placing first connectors through aligned through-openings between the proximal end of the first ballast and the first end of the rail (or first flange); and (ii) placing second connectors through aligned through-openings between the proximal end of the second ballast and the second end of the rail (or second flange).

(52) In one aspect, the method comprises determining which of the through-openings of the first and second flanges that the connectors are to be placed. This determines the relative angle between the weighted bodies of the ballasts and the rail and supported charges. In this way, when the first and second ballasts and the connected rail rotate within a horizontal portion of a wellbore, the charges can be fired downhole at a desired (or pre-selected) orientation.

(53) In one aspect, the charges may be placed within respective receptacles along the frame to fire at orientations of any of 0 degrees, 90 degrees, 180 degrees, or 270 degrees within a horizontal wellbore, depending on the direction in which the charges are inserted into the respective receptacles and the angle at which the ballasts are secured to the rail.

(54) In one aspect, a first electrically conductive contact pin resides within the bore of the first tandem sub. Similarly, a second electrically conductive contact pin resides within the bore of the second tandem sub. A detonator resides within the gun barrel housing adjacent the rail and is in electrical communication with the first and/or second contact pins.

(55) The method may also include the steps of: running the perforating gun assembly into a wellbore at the end of an electric wireline; pumping the perforating gun assembly into a horizontal portion of the wellbore; and allowing the weighted body of the first ballast and the weighted body of the second body to rotate into a downward position, thereby placing the shaped charges into the pre-selected orientation relative to the wellbore.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the present inventions can be better understood, certain illustrations, charts, and/or flow charts are appended hereto. It is to be noted, however, that the drawings illustrate only selected embodiments of the inventions and are therefore not to be considered limiting of scope, for the inventions may admit to other equally effective embodiments and applications.

(2) FIG. 1 is a side, cross-sectional view of a wellbore, in one embodiment. The wellbore has been completed with an elongated horizontal section. A perforating gun assembly is shown having been pumped into the horizontal section, or leg, at the end of an e-line.

(3) FIG. 2 is a side view of an illustrative string of gun barrels forming a perforating gun assembly. The perforating gun assembly represents a series of perforating guns having been threadedly connected end-to-end. Tandem subs are shown between gun barrel housings of the perforating guns, providing the threaded connections.

(4) FIG. 3A is a first perspective view of a tandem sub as may be used in the string of gun barrels of FIG. 2A.

(5) FIG. 3B is a second perspective view of the tandem sub of FIG. 3A.

(6) FIG. 4A is a perspective view of a gun barrel housing as may be used in the perforating gun assembly of the present invention, in one embodiment. The gun barrel housing is straddled by opposing tandem subs.

(7) FIG. 4B is a perspective view of the gun barrel housing of FIG. 4A. The tandem subs have been removed for illustrative purposes.

(8) FIG. 5A is a cut-away view of the gun barrel housing of FIG. 4A. Tandem subs are again seen at opposing ends of the gun barrel housing. Various components of a perforating gun assembly are seen within the gun barrel housing including a rail for supporting shaped charges.

(9) FIG. 5B is an enlarged, cross-sectional view of an end of the perforating gun assembly of FIG. 5A. A contact pin is seen extending through the tandem sub. In addition, a bearing connection is shown between the tandem sub and a ballast.

(10) FIG. 6A is a first perspective view of rail system which resides within the gun barrel housing

of FIGS. 4B and 5A. A rail is seen between opposing ballasts. In this view, each shot is to be oriented at an orientation of 0 degrees.

(11) FIG. 6B is a second perspective view of the rail system of FIG. 6A. In this view, each shot is to be oriented 90 degrees.

(12) FIG. 6C is a third perspective view of the rail system of FIG. 6A. In this view, one shot is oriented at 0 degrees while a second shot is oriented at 180 degrees. Tandem subs are shown at the distal ends of respective ballasts.

(13) FIG. 7 is a perspective view of a bulkhead. The bulkhead houses an elongated contact pin. Banana clips are shown at opposing ends of the contact pin.

(14) FIG. 8A is a perspective view of the rail used in the perforating gun assembly of FIG. 5A. The rail is dimensioned to hold a series of charge jackets, which in turn secure respective charges.

(15) FIG. 8B is a perspective view of a charge jacket as may be used to secure a charge within the rail of FIG. 8A.

(16) FIG. 9A is a perspective view of an eccentric weighted ballast. The ballast is designed to be secured to an end of the rail of FIG. 8A.

(17) FIG. 9B is a perspective view of a cover. The cover is intended to be fitted over the weighted ballast of FIG. 9A. The cover protects wires that extend from the contact pin down into the gun barrel housing along with a detonator and an addressable switch.

(18) FIG. 10 is a cross-sectional view of the tandem sub of the present invention, in one embodiment.

(19) FIG. 11 is a perspective view of a threaded connector. The threaded connector is used to secure the ballast to an end of the rail.

(20) FIG. 12A is an enlarged perspective view of a bearing member. The bearing member provides relative rotational movement between a ballast and an associated tandem sub.

(21) FIG. 12B is a second perspective view of the bearing member of FIG. 12A. Opposing ends of the bearing member are more clearly seen. The bearing member is seen from an end opposite that of FIG. 12A.

(22) FIG. 12C is a perspective view of a tandem sub. Bearing members from FIGS. 12A and 12B are seen in exploded apart relation on opposing ends of the tandem sub.

(23) FIG. 13A is a perspective view of a detonator. Extending from the detonator is a pair of wires.

(24) FIG. 13B is a perspective view of a connector housing. The connector housing is designed to secure a female banana plug terminal along the rail of FIG. 8A. A wire is seen extending from the connector housing, which is crimped or otherwise connected to the banana plug terminal.

(25) FIGS. 14A and 14B together present a single flow chart. The flow chart provides steps for a method of orienting shots during a formation fracturing operation.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

Definitions

(26) For purposes of the present application, it will be understood that the term “hydrocarbon” refers to an organic compound that includes primarily, if not exclusively, the elements hydrogen and carbon. Hydrocarbons may also include other elements, such as, but not limited to, halogens, metallic elements, nitrogen, carbon dioxide, and/or sulfuric components such as hydrogen sulfide.

(27) As used herein, the terms “produced fluids,” “reservoir fluids,” and “production fluids” refer to liquids and/or gases removed from a subsurface formation, including, for example, an organic-rich rock formation. Produced fluids may include both hydrocarbon fluids and non-hydrocarbon fluids. Production fluids may include, but are not limited to, oil, natural gas, pyrolyzed shale oil, synthesis gas, a pyrolysis product of coal, nitrogen, carbon dioxide, hydrogen sulfide and brine.

(28) As used herein, the term “fluid” refers to gases, liquids, and combinations of gases and liquids, as well as to combinations of gases and solids, combinations of liquids and solids, and combinations of gases, liquids, and solids as a slurry.

(29) As used herein, the term “surface” refers to a location on the earth's surface. The surface may

be a land surface or a water surface.

(30) As used herein, the term “subsurface” refers to geologic strata occurring below the earth's surface.

(31) As used herein, the term “formation” refers to any definable subsurface region regardless of size. The formation may contain one or more hydrocarbon-containing layers, one or more non-hydrocarbon containing layers, an overburden, and/or an underburden of any geologic formation. A formation can refer to a single set of related geologic strata of a specific rock type, or to a set of geologic strata of different rock types that contribute to or are encountered in, for example, without limitation: (i) the creation, generation, and/or entrapment of hydrocarbons or minerals; and (ii) the execution of processes used to extract hydrocarbons or minerals from the subsurface region.

(32) As used herein, the term “wellbore” refers to a hole in the subsurface made by drilling or insertion of a conduit into the subsurface. A wellbore may have a substantially circular cross-section, or other cross-sectional shape. The term “well,” when referring to an opening in the formation, may be used interchangeably with the term “wellbore.” A wellbore may be formed for the purpose of producing hydrocarbon fluids. Alternatively, a wellbore may be completed for the purpose of producing steam in connection with a geothermal project.

(33) The terms “upstream” and “downstream” may be used to indicate the relative position of tools or components within a wellbore.

(34) As used herein, the term “sub” generally refers to a cylindrical body. The sub may have opposing threaded ends and is used to connect tubular bodies in series.

(35) Reference herein to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with an embodiment is included in at least one embodiment of the subject matter disclosed. Thus, the appearance of the phrases “in one embodiment” or “in an embodiment” in various places throughout the specification is not necessarily referring to the same embodiment.

Description of Selected Specific Embodiments

(36) The following description of the embodiments refers to the accompanying drawings. The same reference numbers in different drawings identify the same or similar elements.

(37) The following detailed description does not limit the invention. Instead, the scope of the invention is defined by the appended claims.

(38) FIG. 3A is a first perspective view of an illustrative tandem sub **225**. FIG. 3B is a second perspective view of the same tandem sub **225**. The tandem sub **225** will be described with reference to FIGS. 3A and 3B together.

(39) The tandem sub **225** defines a short tubular body having a first end **202** and a second opposing end **204**. The tandem sub **225** may be, for example, 1.00 inches to 5.5 inches in length, with the two ends **202**, **204** being mirror images of one another. Preferably, the tubular body forming the tandem sub **225** is portless.

(40) The tandem sub **225** includes externally machined threads **206**. The threads **206** are male threads dimensioned to mate with female threaded ends of a gun barrel housing. A gun barrel housing is shown at **410** in FIG. 4B. The tandem sub **225** is preferably dimensioned in accordance with standard 3 $\frac{1}{8}$ ” gun components. This allows the tandem sub **225** to be threadedly connected in series with perforating guns from any American vendor, e.g., Geo-Dynamics® and Titan®.

(41) Intermediate the length of the tandem sub **225** and between the threads **206** is a shoulder **216**. The shoulder **216** serves as a stop member as the tandem sub **225** is screwed into an end **412** or **414** of a gun barrel housing **410**. Stated another way, the shoulder **216** serves as a limiter to the tightening of the tandem sub **225** onto the gun barrel housing **410**. Optionally, grooves **217** are formed equi-radially around the shoulder **216**. The grooves **217** removably engage with a tool (not shown) used for applying a rotational force to the tandem sub **225** without harming the rugosity of the shoulder **216**. U.S. Pat. No. 11,559,875 describes a socket driver that may be used to connect the tandem sub **225** to a gun barrel housing **212** or **410**. The '875 patent is titled “Socket Driver, and

Method of Connecting Perforating Guns” and is incorporated herein by reference in its entirety.

(42) The tandem sub **225** includes a central bore **235**. As will be described in greater detail below, the bore **235** is dimensioned to receive a bulkhead (seen at **475** of FIG. 5B). The bulkhead **475**, in turn, supports a contact pin (shown at **550**, also in FIG. 5B). An upstream end of the contact pin **550** is seen at **552** in FIG. 3B.

(43) The contact pin **550** defines an elongated body that is fabricated from brass or a metal alloy comprised at least partially of brass. Thus, the contact pin **550** is electrically conductive.

(44) Circular grooves **212** are formed along the tandem sub **225**. Specifically, a pair of circular grooves **212** is provided on opposing sides of the shoulder **216**. The grooves **212** are dimensioned and configured to receive respective O-rings (not shown). The O-rings preferably define elastomeric seals that closely fit between an outer diameter of the tandem sub **225** and an inner diameter of the surrounding gun barrel housing **410**. The O-rings provide a pressure seal for the gun barrel housing **410** when in the wellbore and prior to charges **540'**, **540''** being detonated.

(45) It is noted that the large O-rings on the O.D. of the tandem sub **225** protect the interior of the barrel (charges, wires, etc.) from wellbore fluid/pressure. Once the charges go off, the interior of the barrel floods. The bulkhead **475** and O-rings on the bulkhead **475** protect the upstream gun from flooding when the gun below is flooded from firing.

(46) FIG. 4A is a perspective view of a gun barrel housing **410**. The gun barrel housing **410** defines a tubular body **420**, preferably fabricated from steel. The tubular body **420** has a first end **412**, and a second end **414** opposite the first end **412**. In the view of FIG. 4A, tandem subs **225** have been screwed into each of the opposing ends **412**, **414** of the gun barrel housing **410**.

(47) The gun barrel housing **410** is dimensioned to house components of any known perforating gun. Such components include a detonator and a detonator cord (shown in FIG. 5A at **560** and **545**.) In addition, the gun barrel housing **410** will house a set of explosive charges **540'**, **540''** used to form perforations into the surrounding gun barrel housing **410** and extending out into the surrounding formation **115**. Uniquely, and as described further below in connection with FIGS. 5A, 5B, 6A, 6B, and 6C, the gun barrel housing **410** does not contain a carrier tube; instead, a novel rail system **600** is employed to support the individual charges **540'**, **540''**. The charges **540'**, **540''** are detonated by the detonator cord **545** at a pre-determined orientation.

(48) FIG. 4B is a perspective view of the gun barrel housing **410** of FIG. 4A. Here, the tandem subs **225** and internal components have been removed for illustrative purposes. It is understood that a series of gun barrel housings **410** will be threadedly connected together using the tandem subs **225** of FIG. 4A. Each gun barrel housing **410** will contain a means for transmitting a detonation signal along with electronics for processing those signals.

(49) It is noted that in FIGS. 4A and 4B, the gun barrel housing **410** has both the first end **412** and the second end **414**. When placed in a wellbore **100**, the first end **412** represents an upstream end, while the second end **414** represents a downstream end. It is understood that in “oil patch” parlance, a left end of a tool represents the upstream end while a right end of a tool represents the downstream end. In practice, a perforating gun string will include a series of perforating gun assemblies (shown at **400** in FIG. 5A) each of which may be between 8 inches and 3 feet in length.

(50) FIG. 5A is a cut-away view of the gun barrel housing **410** of FIG. 4A. Tandem subs **225** are again seen at opposing ends **412**, **414** of the gun barrel housing **410**. In addition, contact pins **550** are visible within the respective tandem subs **225**. The tandem subs **225** are threaded into the bore **415** of the gun barrel housing **410**. A distal end **552** of each contact pin **550** extends into a first conical opening **270** of the corresponding tandem sub **225**. At the same time, a proximal end **554** of each contact pin **550** extends through a second conical opening **270''** of the tandem sub **225**, and then into the bore **415** of the gun barrel housing **410**. Together, internal components contained within the gun barrel housing **410** comprise elements of the perforating gun assembly **400**.

(51) FIG. 5B is an enlarged view of the end **412** of the gun barrel housing **410** of FIG. 5A. Here, one of the contact pins **550** is more visible. Particularly, it can be seen that the distal end **552** of the

contact pin **550** terminates within the first conical (or angled) openings **270'** of the tandem sub **225**. As shown more fully in FIG. 7, the distal end **552** of the contact pin **550** may optionally support a banana terminal **556**. The banana terminal **556** is configured to provide electrical communication with an upstream signal transmission wire (not shown). It is understood that the signal transmission wire travels down from an upstream gun barrel housing and is itself in electrical communication with the e-line **240**.

(52) It is believed that the banana terminal **556** arrangement provides better electrical connectivity than the so-called "pin-on-seat" arrangement used in many perforating gun systems today. The pin-on-seat arrangement relies on a spring-loaded connection that may not always function properly. In the current electrical system, a small brass female banana plug **558** resides within a connector housing **580**. A wire **585** is then crimped or soldered to the banana plug terminal **558**.

(53) The contact pin **550** resides within a bulkhead **475**. The contact pin **550** includes an enlarged outer diameter portion **555**. The enlarged outer diameter portion **555** is arranged with a series of grooves machined into its profile. The grooves (not numbered) cooperate with mating grooves (also not numbered) along an inner diameter of the bulkhead **475**. Such an arrangement is described in greater detail in co-owned U.S. Pat. No. 10,914,145 which is incorporated herein in its entirety by reference.

(54) Skipping over to FIG. 7, FIG. 7 is a perspective view of the bulkhead **475**. The bulkhead **475** houses the elongated contact pin **550**. The contact pin **550** can be seen extending out of opposite ends of the bulkhead **475**. Banana plugs **556**, **558** are shown at opposing ends **552**, **554** of the contact pin **550**. Specifically, banana plug **556** is at the upstream end **552** while banana plug **558** is at the downstream end **554**. It is noted that the bulkhead **475** of FIG. 7 is presented in a reverse orientation from how it is presented in FIG. 5B. However, the same bulkhead **475** is shown in both FIGS. 7 and 5B.

(55) The bulkhead **475** includes a portion **478** having an enlarged inner diameter. This portion **478** accommodates the corresponding enlarged outer diameter portion **555** of the contact pin **550**. O-rings **476** are shown along the outer diameter of the enlarged inner diameter portion **478**. The O-rings **476** and the enlarged inner diameter portion **478** limit the transmission of fluid and pressure when a downstream perforating gun is detonated.

(56) Returning to FIG. 5B, a castle nut **275** is placed around the contact pin **550** at the downstream end (this is end **554** that extends into the bore **415** of the gun barrel housing **410** as seen in FIG. 5A). The threaded castle nut **275** holds the bulkhead **475** in the tandem sub **225** and in turn holds the contact pin **550** within the bore of the bulkhead **475** during assembly of the perforating gun assembly **400**.

(57) Below the castle nut **275** is a connector housing **580**. The connector housing **580** secures a wire **585** extending from the female banana plug terminal **558** and into the gun barrel housing **410**. The wire **585** serves as a signal transmission wire, transmitting detonation signals to perforating gun assemblies **400** downhole, from gun-to-gun. Those of ordinary skill in the art will understand that addressable switches **610** residing along the gun barrel housings **418** will watch for a detonation signal addressed to a particular perforating gun assembly **400**, causing a detonation signal to be sent into a detonator wire.

(58) It is noted that wire **585** connects to the addressable switch **610**. The addressable switch **610**, in turn, connects to a detonator **560** as well as a wire on the female banana plug at an opposing end of the rail/ballast assembly.

(59) In FIG. 5B, a detonator wire is shown at **565**. The detonator wire **565** extends into the detonator **560**. In actuality, the detonator wire **565** is actually a pair of wires **565P**, **565N** shown in FIG. 13A. When the detonator **560** is activated, a controlled explosion takes place that ignites a detonator cord (shown at **545** in FIG. 5A). The detonator cord **545**, in turn, ignites sets of the shaped charges **540'**, **540''** within the gun barrel housing **410**.

(60) It is observed that in FIG. 5A, three shaped charges **540'**, **540''** are shown. Two outer charges

540' are oriented in a first direction while a middle charge **540"** is oriented in a direction that is rotated by 180 degrees relative to the charges **540'**. Directing charges within a gun barrel housing **410** in two different orientations allows the operator to shoot charges into the surrounding formation **115** in opposite directions.

(61) In the typical carrier tube arrangement, charges are spaced apart radially and longitudinally along a tubular body. This allows shots to be fired in all radial directions through the production casing **150**. However, in the arrangement of FIGS. 5A and 6A, charges **540'**, **540"** are intentionally aligned at nominally 180 degree relation. This allows the operator to shoot charges horizontally from the wellbore **100** once the perforating gun assemblies **400** are in a horizontal position and the rails (see rail **630** in FIG. 6B) rotate into place.

(62) As noted above, it is also desirable to be able to control the orientation at which the shots are fired from the perforating gun assembly **400**. To enable the operator to direct the orientation of the shots fired from the charges **540'**, **540"**, a novel rail system is provided herein. The rail system is shown at **600** and described in connection with FIGS. 6A and 6B. Generally, the rail system **600** comprises a rail **630**, and ballast members **620** which reside at opposing ends of the rail **630**. As will be discussed further below, the rail system **600** allows the charges **540'**, **540"** to freely rotate within the horizontal portion **156** of the wellbore **100**, placing the charges **540'**, **540"** into a predetermined orientation.

(63) FIG. 6A is a first perspective view of the rail system **600**. The rail system **600** is configured to reside within the gun barrel housing **410** of FIGS. 4B and 5A. In this view, each shot **540** is oriented at 0 degrees. (Note that for illustrative purposes only one of three shots is shown in FIG. 6A.)

(64) FIG. 8A is a perspective view of the rail **630** used in the perforating gun assembly **400** of FIG. 5A. The rail **630** defines an elongated frame that is dimensioned to hold a series of charge jackets (shown at **670** in FIGS. 6A and 8B). The charge jackets **670**, in turn, secure respective charges **540**. Preferably, the rail **630** is fabricated from either cast aluminum or an aluminum alloy so as to be conductive along a grounding path. Alternatively, the rail **630** may be fabricated from a rigid polymeric material.

(65) The rail **630** has a first end **632** and a second end **634** opposite the first end **632**. Each end **632**, **634** represents a circular member, or flange, having a plurality of equi-radially positioned through-openings **612**. Preferably, each end **632**, **634** comprises at least four, and more preferably eight, through-openings **612**.

(66) The through-openings **612** are dimensioned and configured to receive a bolt or screw or other securing member (shown at **629** in FIG. 9A). The securing members **629**, in turn, connect the ends **632**, **634** to respective ballasts (shown at **620** in FIG. 6A). In this respect, each ballast **620** contains through-openings (shown at **622** in FIG. 6A) which may be aligned with through-openings **612** of a respective end **632**, **634** of the rail **630** for connection.

(67) It is understood that the first **632** and second **634** flanges may be located on the distal ends of the two ballasts **620**. In this case, the through-openings **622** would be positioned on opposing ends of the rail **630**. Thus, the use of a flange for connecting the rail **630** to the ballasts **620** is not limited to the location of the flanges. In lieu of flanges and through-openings **622**, a clocking mechanism could also be used to adjust the relative rotational positions of the ballasts **620** and the intermediate rail **630**.

(68) Each of the ends **632**, **634** of the rail **630** comprises a central opening **615**. The central openings **615** are dimensioned to receive the detonator cord **545**. This allows the detonator cord **545** to pass from the detonator **560**, through a central opening **615**, into the rail **630**, and on to the respective charges **540'**, **540"**.

(69) The rail **630** further comprises a series of receptacles **635**. The receptacles **635** are preferably equi-distantly spaced along the length of the rail **630**. The receptacles **635** are uniquely configured to receive respective charge jackets **670**. In one aspect, each receptacle **635** has radial edges **631** to

accommodate the typically-circular shaped charges **540**.

(70) FIG. **8B** is a perspective view of an illustrative charge jacket **670**. The charge jacket **670** is preferably fabricated from a flexible polymeric material. The charge jacket **670** is dimensioned to receive and hold any off-the-shelf shaped charge **540'** or **540''**. The flexible polymeric material of the charge jacket **670** allows the charge jacket **670** to hold the respective shaped charge **540'** or **540''** through a snap-fit arrangement.

(71) The charge jacket **670** comprises a series of side rails **672**. The side rails **672** are placed along opposing sides of the charge jacket **670**. The side rails **672** help to support the shaped charges **540'** or **540''**. The side rails **672** are also arranged to slidably engage into cooperating grooves **632** adjacent each receptacle **635** in the rail **630**. Thus, the side rails **672** play a role in holding the charges **540'**, **540''** in place along the rail **630**.

(72) Returning to FIG. **6A**, it can be seen that the charge jackets **670** have been slid into place along each of the three receptacles **635** along the rail **630**. Each of the charge jackets **670** is oriented in the same direction, to wit, 90 degrees from horizontal. This will cause the ensuing shots to be fired in the wellbore vertically.

(73) As noted above, in FIG. **6A** a single charge **540** is shown placed into a charge jacket **670**. Upper transverse members (shown at **674** in FIG. **8B**) hold the charge **540** in place when the charge **540** is placed within (or snapped into) the charge jacket **670**. At the same time, lower transverse members (shown at **676** in FIG. **8B**) hold the charge **540** in place when the charge jacket **670** is lowered into a receptacle **635** of the rail **630**.

(74) Beneficially, the charge jacket **670** is fabricated from a compliant (or elastic) material, allowing it to be deformed as it is inserted into respective cooperating grooves **632** along the rail **630**, and then return to its original shape to remain within the receptacle **635** of the rail **630**. Similarly, the charge jacket **670** may be stretched to capture the shaped charge **540**, and then rebound to its original shape to hold the charge **540** securely in place.

(75) As noted, FIG. **6A** also shows two ballasts **620**. One ballast **620** is secured to the first end **632** of the rail **630** while the other ballast **620** is secured to the second end **634** of the rail **630**. Each ballast **620** employs an identical semi-circular profile. Preferably, each ballast **620** is fabricated from cast zinc and is part of the ground path for the perforating gun assembly.

(76) FIG. **9A** is a perspective view of the ballast **620**, in one embodiment. As the name implies, the ballast **620** comprises a weighted body **624**. The weighted body **624** has a first end **621** and a second, opposing end **623**. At the first end **621**, the weighted body **624** supports a tubular member **625**. The tubular member **625** is dimensioned to receive a bearing member (shown at **660** in FIGS. **6A** and **6B**). Specifically, the tubular member **625** receives the bearing member **660** around its outer diameter. Longitudinal movement of the bearing member **660** along the tubular member **625** is restricted by a shoulder **627** above the weighted member **624**.

(77) The second end **623** of the weighted body **624** comprises a pair of wings **626**. Each wing **626** contains the through-opening **622** referenced above. When the through-openings **622** are aligned with through-openings **612** of the rail **630**, a threaded connector **629** is placed through the aligned through-openings **622**, **612**. The threaded connectors **629** hold the ballast **620** in place relative to the rail **630** and supported charges **540'**, **540''**.

(78) In an alternative arrangement, a pair of press-fit pins (not shown) are run into the aluminum flanges **632**, **634** of the rail **630** to secure wings **626** to the opposing flanges **632**, **634**.

(79) Intermediate the first **621** and the second end **623** of the ballast **620** is a channel **628**.

Consistent with the semi-circular profile of the ballast **620**, the channel **628** is open. The channel **628** is dimensioned to receive a signal transmission wire, such as wire **585** of FIG. **5B**, extending down from an upstream perforating gun **410**. Optionally, the channel **628** may secure the detonator **560**. The channel **628** may also be used to contain an addressable switch **610** as shown in FIG. **5B**

(80) FIG. **11** is a perspective view of a threaded connector **629**, in one arrangement. The connector **629** comprises a threaded bolt and is dimensioned to extend through the aligned through-openings

622, 612. The connector **629** is screwed into the aligned through-openings **622, 612** from the direction of the ballast **630**.

(81) In a preferred embodiment, eight through-openings **622, 612** are provided on the end flanges **632, 634** to allow the operator to select a desired orientation for the charges **540', 540''**. A head **1129** serves as a stop member for the connector **629** that prevents the connector **629** from moving completely through the through-openings **622, 612** during assembly. Accordingly, the shoulder **1129** permits the connector **629** to thread into the through-openings **622, 612**, secure the ballast members **620** to the rail **630**, and prevent over-insertion of the connector **629** during assembly.

(82) FIG. 12A is an enlarged perspective view of a bearing member **660**. The bearing member **660** has a frusto-conical profile, with a central bore **665**. The central bore **665** is dimensioned to closely receive the tubular support member **625** at the first (or distal) end **621** of the ballast **620**. FIG. 12B is a second perspective view of the bearing member **660** of FIG. 12A, seen from an end opposite that of FIG. 12A.

(83) The bearing member **660** includes a plurality of individual roller bearings **662**. The roller bearings **662** are secured within a housing **664**, forming a race for the bearings **662**. The bearings **662** interface with the second conical opening **270''** of the tandem sub **225**, meaning that the roller bearings **662** permit the bearing member **660** to rotate within the second conical opening **270''**. In this way, the bearing member **660** provides relative rotational movement between the tandem sub **225** and the ballast **620** (and its connected rail **630** and supported shaped charges **540', 540''**) within the gun barrel housing **410**.

(84) FIG. 12C is a third perspective view of the bearing member **660** of FIG. 12A. Here, two opposing bearing members **660** are shown in exploded apart relation from a tandem sub **225**. Each bearing member **660** is dimensioned and configured to slide into the bore **235** of the tandem sub **225** and engage a conical surface **270', 270''**. (The conical surfaces **270', 270''** are shown in FIG. 10, discussed below.)

(85) Stated another way, each bearing member **660** is designed to engage with a tandem sub **225**, wherein the tandem sub **225** comprises a first end **202**, a second end **204** opposite the first end **202**, connection threads **206** placed at each of the first **202** and second **204** ends, an internal bore **235** within the tandem sub **225** extending from the first end **202** to the second end **202**. As shown in FIG. 10, the tandem sub **225** includes a pair of tapered shoulders **270', 270''** along the inner bore **235**.

(86) Each bearing member **660** comprises a first end, or shoulder **661**, defining a first outer diameter. In addition, each bearing member **660** comprises a second end, or shoulder **663**, defining a second outer diameter. The second outer diameter is smaller than the first outer diameter creating the frusto-conical profile. The angle of the frusto-conical profile will match the angle of the tapered shoulder **270' or 270''**.

(87) A tubular support body **668** extends between the first shoulder **661** and the second shoulder **663**. The support body **668** is dimensioned to receive the post **625** of the ballast **620** while supporting the bearing member **660**. At the same time, the support body **668** serves as a race.

(88) A plurality of roller bearings **662** is disposed around the support body **668** intermediate the first shoulder **661** and the second shoulder **663**. The plurality of roller bearings **662** is configured to bearingly engage the tapered shoulder **270'** along the inner bore **235** of the tandem sub **225**. The roller bearings **662** are contained within the housing **664**. The housing **664** serves as a cage for the roller bearings **662** over the race **668**. At the same time, the housing **664** provides windows **667** through which the respective roller bearings **662** extend.

(89) The position of the bearing member **660** is shown in FIG. 6B. FIG. 6B is a second perspective view of the rail system **600** of FIG. 6A. The bearing member **660** is seen secured over the post **625** at the first end **621** of a ballast **620**. In turn, a flange **680** at the end of the post **625**. The flange **680** is associated with a connector housing (shown at **580** in FIG. 13B). As discussed below, the connector housing **580** serves multiple purposes including receiving the female banana plug

terminal **558**, holding the bearing **660** in place on the ballast post **625**, and supporting the wire **585** that is crimped or soldered onto the female banana plug terminal **558**.

(90) Of interest, the bearings **662** interface with only the ID of the tandem subs **225**; they do not interface with the ID of the surrounding gun barrel housing **410**. This allows for the use of smaller bearings **662**, which in turn allows for reduced cost and lower friction.

(91) The frusto-conical, or tapered, roller bearings **662** have a high side load rating and better durability. The tapered arrangement for the roller bearings **662** also provides a better electrical/ground connection with a bigger surface area of contact to improve electrical communication. Those of ordinary skill in the art may understand that the tandem subs **225** are cleaner and offer more reliable ground surfaces than the ID's of traditional gun barrels.

(92) In one embodiment, roller bearings are integral to the tapered surfaces **270'**, **270''** along the inner bore **235** of the tandem sub **225**. In this instance, the housing **664**, or cage, is built into the inner bore **235**. Each ballast **620** would employ a matching angle tapered surface at its first end **621**. For example, the post **625** could be modified to include a tapered, or frusto-conical, surface.

(93) In one aspect, the tapered shoulder **270'** or **270''** represents a short sub, e.g., one to three inches, that is threadedly connected to an end of the tandem sub **225**. In this instance, the short sub has an outer diameter that is the same as the outer diameter of the tandem sub **225**, but employs the tapered shoulder **270'** or **270''** along its inner bore. The bearing member **660** engages the tapered shoulder of the end cap. For purposes of the claims herein, the end cap essentially becomes a part of the tandem sub **225** once it is screwed on.

(94) It is observed that in FIG. **6B**, the rail **630** and supported charge jackets **670** have been rotated 90 degrees relative to the view of FIG. **6A**. This is because the first end **632** of the rail **630** has been rotated 90 degrees before being secured to the second end **623** of the ballast **620**. The result is that the shaped charge **540** is now oriented 90 degrees from vertical. This is a preferred orientation of the charge **540** as it ensures that shots will be fired parallel to a plane of completion of the horizontal portion **156** of the wellbore **100**. This also allows the operator to take advantage of the plane of least resistance, or "least principal stress," within the formation **115** to maximize fracture formation away from the wellbore **100** and within the stress field.

(95) FIG. **6C** is a third perspective view of the rail system **600** of FIG. **6A**. In this view, one shot **540'** is oriented at 0 degrees while another shot **540''** is oriented at 180 degrees. Thus, the shaped charges **540'**, **540''** may be installed along the rail **630** in either an "up" or a "down" direction. "Up" may be considered a "0 degree" orientation while down may be considered a "180 degree" orientation. This arrangement may be desirable to help minimize fracture communication with a so-called parent wellbore in the field. In this respect, fractures are led to propagate up and down rather than sideways.

(96) In the arrangement of FIGS. **6A**, **6B** and **6C**, each weighted ballast **620** has an eccentric profile. The weight bodies **624** have rotated into a downward position. Knowing that this will take place when the perforating gun assembly **400** has been pumped into a horizontal leg **156**, the operator can pre-select an orientation (reflecting both a direction and an angle relative to the central axis of the wellbore) at which shots will be fired into the formation **115**.

(97) FIG. **9B** is a perspective view of a cover **690**. The cover **690** is intended to be fitted over the weighted body **624** of FIG. **9A**. Specifically, the cover **690** fits over the open channel **628** to protect wires that extend from the contact pin **550**, or wires (see in FIG. **13A** at **565N**, **565P**) that extend from the detonator **560**, an addressable switch **610**, and any other components of the perforating gun assembly **400**. The cover **690** may be fabricated from a metal or from a durable plastic.

(98) FIG. **10** is a cross-sectional view of the tandem sub **225** of the present invention, in one embodiment. As can be seen, the tandem sub **225** has a bore **235** that extends from the first end **202** to the second end **204**. A first tapered shoulder **270'** is provided along the bore **235** (or inner diameter) at the first end **202** of the tandem sub **225**. Similarly, a second tapered shoulder **270''** is provided along the bore **235** at the second end **204** of the tandem sub **225**.

(99) FIG. 13A is a perspective view of a detonator **560**. The detonator **560** defines a small aluminum housing having a resistor (not seen) inside. The detonator **560** receives electrical energy from the surface **105** and through a pair of wires. These represent a positive wire **565P** and a negative wire **565N**. Current supplied by the electrical energy from the surface **105** heats the resistor within the detonator **560**.

(100) FIG. 13B is a perspective view of the connector housing **580** referenced above. The connector housing **580** is designed to secure the bearing member **660** onto the post **625**. Specifically, the connector housing **580** snaps into the bore of the post **625**. In addition, the connector housing **580** receives the female banana plug **558**. The connector housing **580** is fabricated from a non-conductive composite material.

(101) In operation, the operator will send an electrical signal from the surface **105**, down the electric wireline **240**, through the body of the contact pin **550**, through the female banana plug **558**, to the addressable switch **610**, into one of the detonator wires (such as **565P**), travel into the detonator **560** inside of the gun barrel housing **410**, and then return to ground through another one of the detonator wires (such as **565N**). The detonator **560** is packed with an explosive such as RDX. When current is run through the detonator **560**, a small explosion is set off by the electrically heated resistor. This small explosion then sets off the detonator cord **545** along the selected gun barrel housing **410**.

(102) Of interest, the second detonator wire is connected to the ballast **620**, which is fabricated from a conductive material. The ballast **620**, in turn is connected to the bearing member **660**. The bearing member **660** is in electrical communication with the tandem sub **225** and then the gun body **410** through thread **206**. In this way, the bearing member **660** plays a role in grounding the perforating gun assembly.

(103) In practice, a gun barrel housing **410** may have a tandem sub **225**, a bearing member **660**, a ballast **620** and a rail **630** (or explosive charge holder) at each of its first **412** and second **414** ends. In this instance, a grounding wire (such as wire **565N**) is in electrical communication with the gun barrel housing **410** through the bearing members **660** at both ends **412**, **414**. The benefit is that each end (**412** may be the upstream end and **414** may be the downstream end) is part of the same ground circuit. In the unlikely event that one ground fails (such as through dirt on the roller bearings **662** of one bearing member **660** or because a ground wire **565N** becomes disengaged), the perforating gun assembly is still grounded through the other bearing member **660**. In other words, the ground path is redundantly designed.

(104) In another embodiment, a grounding wire extends through the inner bore **415** of the gun barrel housing **410** from bearing member **660** to bearing member **660**. More specifically, the wire would travel from connector housing **580**/banana plug terminal **558** to connector housing **580**/banana plug terminal **558**.

(105) The bearing connection (including bearing members **662**) provided between the tandem sub **225** and the ballast **620** also provides the operator with at least some measure of control concerning the orientation (or altitude) of charges **540'**, **540''** fired into the surrounding formation **115**. In this respect, once the perforating gun assembly **400** (that is, the tandem subs **225**, the gun barrel housing **410**, and the rail system **600** within the gun barrel housing **410**) enters the horizontal leg **156** of the wellbore **110**, the weighted bodies **624** of the ballasts **620** will roll downward. Because the ballasts **620** are secured to the ends **632**, **634** of the rail **630**, the rail **630** and its supported charges **540'**, **540''** will rotate with the ballasts **620**. The rail **630** is symmetrically balanced to be "neutral," allowing the conductive ballasts **620** to fully dictate the direction the charges **540'**, **540''** will face.

(106) Beneficially, no charge tubes are required for placement within the gun barrel housing **410**. In this respect, the rail system **600** supports the charges **540'**, **540''** along the gun barrel housing **410** rather than a charge tube. In addition, no metal end plates are required at the ends **202**, **204** of the tandem subs **225**. Instead, the ballasts **620** and rail **630** provide structural support for the detonator

560 and associated charges 540', 540".

(107) Based on the tandem subs 225, the eccentric ballasts 620, and the unique bearing connection between the tandem subs 225 and the ballasts 620 within the gun barrel housing 410, a method of orienting shots during a formation fracturing operation is provided herein. FIGS. 14A and 14B together present a single flow chart showing steps for a method 1400 of orienting shots in a perforating gun assembly.

(108) In one embodiment, the method 1400 first comprises providing a gun barrel housing. This is shown in Box 1410 of FIG. 14A. The gun barrel housing may be in accordance with the illustrative gun barrel housing 410 presented in FIG. 4B.

(109) The method 1400 also includes providing a rail. This is seen in Box 1415. The rail is configured to support charges (such as shaped charges) along a frame. Specifically, receptacles are provided along the frame in a preferably equi-distantly spaced array in order to receive the shaped charges. The rail may be in accordance with the rail 630 shown in FIG. 8A.

(110) The method 1400 additionally comprises connecting ballasts to ends of the rail. This is offered in Box 1420. The ballasts are connected using threaded cutting screws that are run into aligned through-openings. In an alternate embodiment, the ballasts may be connected via screws (such as connector 629 of FIG. 11), that are run through the aligned through-openings. Each of the ballasts may be in accordance with the ballast 620 of FIG. 9A.

(111) In operation, the ballasts are connected to opposing ends of the rail. The ballasts are connected such that when the gun barrel housing is pumped into a horizontal leg of a wellbore, the ballasts will be gravitationally rotated into a downward position. This downward rotation of the ballasts will turn the receptacles along the rail within the wellbore at desired orientations.

(112) The method 1400 also includes the optional step of providing charge jackets for the shaped charges. This is provided in Box 1425. Each charge jacket is dimensioned to receive and hold any "off-the-shelf" shaped charge. The charge jackets may be in accordance with charge jacket 670 of FIG. 8B.

(113) The method 1400 also comprises the optional step of determining a location of adjacent wellbores in a hydrocarbon producing field. This is shown at Box 1430. Those of ordinary skill in the art will understand that the hydrocarbon field may, and likely will, comprise numerous wells. In connection with so-called in-fill drilling, new wells are drilled in relative proximity to existing wells. The new well is sometimes referred to as a "child wellbore" or an "offset well," while an existing adjacent well may be referred to as a "parent wellbore."

(114) In this instance, the operator may desire that shots be fired not only horizontally, but also in one direction only, that is, away from a parent wellbore. This assists the service company in generating and propagating fractures in a particular part of the formation to avoid frac hits. Those of ordinary skill in the art will appreciate that frac hits are generally a by-product of in-fill drilling. Frac hits are also, of course, a by-product of tight well spacing, that is, wells spaced in close proximity to one another in the same field. Ultimately, however, frac hits are the result of the operator being unable to control or "direct" the propagation of fractures within the pay-zone. By firing shots away from a direction of a parent well, the likelihood of a frac hit occurring is reduced.

(115) The method 1400 further includes the step of placing the shaped charges into the charge jackets. This is provided at Box 1435. Thereafter, the charge jackets are inserted into receptacles (such as receptacles 635) provided along the rail. This is shown at Box 1440.

(116) It is noted that in lieu of charge jackets, the operator may simply clip the shaped charges onto the receptacles. What is important is that the charge jackets are inserted in select directions. Depending on the placement of the charges along the rail and the angle at which the ballasts are secured to the rail, charges may be shot at a variety of orientations. Examples may be 90-90-90, 270-270-90, 0-90-180, etc.

(117) Beneficially, this allows the operator to orient charges away from a known pressure sink in the formation, such as a pressure sink caused by a parent wellbore in the same field. To facilitate

this, the charge jackets are run into the rail from only one direction, e.g., 0-0-0, 90-90-90, 180-180-180, or 270-270-270. This minimizes the chances of a frac hit occurring during formation fracturing.

(118) Optionally, charge jackets may be placed along the rail in a way that charges will fire vertically, e.g., 0-180-0. This too may minimize the chances of a frac hit occurring during formation fracturing. Of course, the rail may be connected to the ballasts at any angle and the charges may be inserted into the rail receptacles from either direction, providing a high degree of pre-selection for the orientation of the charges.

(119) The method **1400** additionally includes connecting the charges to a detonator cord. This is offered in Box **1445**. The operator then positions the rail and the supported charges within the gun barrel housing. This is seen in Box **1450** of FIG. **14B**.

(120) As discussed above, the novel perforating gun assembly described herein offers a bearing connection within the gun barrel housing. Thus, in Box **1455** the method includes the step of providing a bearing member at an end of each ballast. Each of the bearing members may be in accordance with the bearing member **660** of FIG. **12**. In this respect, the bearing members form a frusto-conical profile, mating with a conical inner diameter section along a tandem sub.

(121) The method **1400** next includes threadedly connecting each end of the gun barrel housing to a tandem sub. This is seen in Box **1460**. Upon connection, the bearing members engage an inner diameter of the respective tandem subs. Together, the tandem subs, the gun barrel housing, the roller bearings (or bearing members), the ballasts, the rail, and the charges form a novel perforating gun assembly.

(122) As an alternative, at least one bearing member is connected to an end plate, which in turn is connected to a gun barrel housing. The end plate comprises a short sub, such as one to three inches, that screws onto the second end of the gun barrel housing. In this case, an opposite end of the end plate may screw into the first end of an adjoining gun barrel housing, or onto a setting tool for a plug, as the case may be.

(123) The method **1400** may also include running the perforating gun assembly into a wellbore. This is provided at Box **1465**. This step is typically done by running the perforating gun assembly into the wellbore at the end of an electric wireline (such as wireline **240** of FIG. **1**). The perforating gun assembly is then pumped into a horizontal portion of the wellbore (such as horizontal leg **156** of FIG. **1**). This is shown at Box **1470**.

(124) Once the perforating gun assembly is positioned within the horizontal wellbore, the ballasts are allowed to rotate into a downward position. This occurs automatically in response to the gravitational force acting on the weighted bodies. This step is shown in Box **1475**. This rotation takes place through the bearing members. The result is that the shaped charges are oriented at a desired angle or direction within a surrounding formation.

(125) It will be appreciated by the petroleum engineer that the size and orientation of a fracture, and the amount of hydraulic pressure needed to part the rock along a fracture plane, are dictated by the formation's in situ stress field. This stress field can be defined by three principal compressive stresses which are oriented perpendicular to one another. These represent a vertical stress, a minimum horizontal stress, and a maximum horizontal stress. The magnitudes and orientations of these three principal stresses are determined by the geomechanics in the region and by the pore pressure, depth, and rock properties.

(126) According to principles of geomechanics, fracture planes will generally form in a direction that is perpendicular to the plane of least principal stress in a rock matrix. Stated more simply, in most wellbores, the rock matrix will part along vertical lines when the horizontal section of a wellbore resides below 3,000 feet, and sometimes as shallow as 2,000 feet, below the surface. In this instance, hydraulic fractures will tend to propagate from the wellbore's perforations in a vertical, elliptical plane perpendicular to the plane of least principal stress. If the orientation of the least principal stress plane is known, the longitudinal axis of the horizontal leg **156** is ideally

oriented parallel to it such that the multiple fracture planes will intersect the wellbore **100** at, or near, orthogonal to the horizontal leg **156** of the wellbore **100**.

(127) The method **1400** may finally include sending an actuation signal down the electric line to initiate charges and to create perforations in a direction that is generally opposite from a direction of the parent wellbore, thus avoiding a frac hit in the hydrocarbon producing field while optimizing the creation of a fracture network within the formation. This may be a part of the step of Box **1475**.

(128) Further, variations of the tool and of methods for using the tool within a wellbore may fall within the spirit of the claims, below. It will be appreciated that the inventions are susceptible to other modifications, variations and changes without departing from the spirit thereof.

Claims

1. A perforating gun assembly, comprising: a gun barrel housing having a first end, a second end opposite the first end, and a bore extending from the first end to the second end of the gun barrel housing; a rail comprising an elongated frame having at least one receptacle, wherein each of the at least one receptacle is configured to receive an explosive charge; a first tandem sub threadedly connected to the first end of the gun barrel housing, and a second tandem sub threadedly connected to the second end of the gun barrel housing, wherein each of the first and second tandem subs comprises a first end, a second end opposite the first end, and a bore extending from the first end to the second end of the respective tandem subs; a first ballast and a second ballast, wherein each of the first and second ballasts comprises an eccentric weighted body, and wherein: the first ballast comprises a proximal end connected to a first end of the rail, and a distal end abutting the first tandem sub; and the second ballast comprises a proximal end connected to a second end of the rail, and a distal end abutting the second tandem sub; a first bearing member supported by the first ballast at the distal end of the first ballast, and a second bearing member supported by the second ballast at the distal end of the second ballast; wherein the first bearing member interfaces with the first tandem sub and the second bearing member interfaces with the second tandem sub, thereby allowing for relative rotation between the rail and the first and second tandem subs.
2. The perforating gun assembly of claim 1, wherein: each of the first tandem sub and the second tandem sub comprises male threaded ends; and the first and second opposing ends of the gun barrel housing each comprises female threads forming a female-by-female tubular body.
3. The perforating gun assembly of claim 1, wherein the rail is configured such that an explosive charge may be received within a respective receptacle from either side of the elongated frame.
4. The perforating gun assembly of claim 3, wherein: a first flange resides at a first end of the rail; a second flange resides at a second end of the rail; the first ballast comprises a proximal end connected to the first flange; and the second ballast comprises a proximal end connected to the second flange.
5. The perforating gun assembly of claim 1, wherein: the at least one receptacle comprises at least three receptacles; the connection between the proximal end of the first ballast and the first end of the rail comprises a first flange having at least four spaced-apart through-openings; the connection between the proximal end of the second ballast and the second end of the rail comprises a second flange having at least four spaced-apart through-openings; the proximal end of the first ballast comprises a pair of through-openings that may be aligned with a selected pair of the through-openings of the first flange; and the proximal end of the second ballast comprises a pair of through-openings that may be aligned with a selected pair of the through-openings of the second flange.
6. The perforating gun assembly of claim 5, wherein the proximal end of the first ballast is connected to the first flange, and the proximal end of the second ballast is connected to the second flange using posts, metal pins, or threaded screws.
7. The perforating gun assembly of claim 1, further comprising: a plurality of explosive charges; and a plurality of charge jackets, wherein each charge jacket is fabricated from a compliant

material that receives a respective explosive charge; and wherein each charge jacket is received within a respective receptacle.

8. The perforating gun assembly of claim 7, wherein: the at least one receptacle comprises at least two receptacles; the at least two receptacles are equi-distantly spaced; the rail further comprises slots associated with each of the receptacles, and each of the charge jackets comprises side rails configured to be received within the slots of a respective receptacle along the frame.

9. The perforating gun assembly of claim 1, wherein: each of the first ballast and the second ballast comprises a support member at its distal end; and each of the first and second bearing members defines (i) a tubular body connected to a respective support member of the first ballast and the second ballast, serving as a race, and (ii) a plurality of roller bearings along the race; such that the roller bearings of the first bearing member interface with an inner diameter along the bore of the first tandem sub, and the roller bearings of the second bearing member interface with an inner diameter along the bore of the second tandem sub.

10. The perforating gun assembly of claim 9, further comprising: a first electrically conductive contact pin residing within the bore of the first tandem sub; a second electrically conductive contact pin residing within the bore of the second tandem sub; and a detonator residing along the first ballast adjacent the rail.

11. The perforating gun assembly of claim 10, wherein the explosive charges may be placed within respective receptacles along the frame to fire at an orientation of any of 0 degrees, 90 degrees, 180 degrees, or 270 degrees within a horizontal wellbore.

12. The perforating gun assembly of claim 11, wherein: each of the first and second ballasts is fabricated from zinc; and the rail is fabricated from aluminum, an aluminum alloy, or a rigid polymeric material.

13. A method of orienting shots in a perforating gun assembly, comprising: providing a perforating gun assembly, wherein the perforating gun assembly comprises: a gun barrel housing having a first end, a second end opposite the first end, and a bore extending from the first end to the second end of the gun barrel housing; a rail comprising an elongated frame having a plurality of receptacles, wherein each receptacle is configured to receive an explosive charge; a first tandem sub and a second tandem sub, with each of the first and second tandem subs having an inner bore; a first ballast and a second ballast, wherein each of the first and second ballasts comprises a weighted body, and wherein: the first ballast comprises a proximal end and a distal end, with the distal end of the first ballast abutting the first tandem sub; and the second ballast also comprises a proximal end and a distal end, with the distal end of the second ballast abutting the second tandem sub; placing an explosive charge within each of the respective receptacles along the frame; threadedly connecting the first tandem sub to the first end of the gun barrel housing, and threadedly connecting the second tandem sub to the second end of the gun barrel housing; and providing a first bearing connection between the first ballast and the first tandem sub along the inner bore of the first tandem sub, and providing a second bearing connection between the second ballast and the second tandem sub along the inner bore of the second tandem sub, such that the first and second ballasts and the connected rail may rotate relative to the first and second tandem subs and connected gun barrel housing.

14. The method of orienting shots of claim 13, wherein the charges may be received within a respective receptacle from either side of the frame.

15. The method of orienting shots of claim 14, further comprising: selecting a direction for each of the explosive charges to be inserted into a respective receptacle along the frame; and connecting the proximal end of the first ballast to the first end of the rail and connecting the proximal end of the second ballast to the second end of the rail, such that the explosive charges will fire at a desired orientation when the first and second ballasts and the connected rail rotate within a horizontal portion of a wellbore.

16. The method of orienting shots of claim 15, wherein: each of the first and second tandem subs

comprises a first end, a second end opposite the first end, and a tapered shoulder along the inner bore at the respective second ends; the perforating gun assembly further comprises a first bearing member supported by the first ballast at the distal end of the first ballast, and a second bearing member supported by the second ballast at the distal end of the second ballast; and the first bearing member interfaces with the tapered shoulder along the inner bore of the first tandem sub, and the second bearing member interfaces with the tapered shoulder along the inner bore of the second tandem sub, thereby allowing for relative rotation between the rail and the first and second tandem subs.

17. The method of orienting shots of claim 15, wherein: each of the first tandem sub and second tandem sub comprises male threaded ends; and the first and second opposing ends of the gun barrel housing each comprises female threads forming a female-by-female tubular body.

18. The method of orienting shots of claim 16, further comprising: placing each of the explosive charges into a respective charge jacket, wherein: each charge jacket is fabricated from a compliant material; the rail further comprises slots associated with each of the receptacles; and each of the charge jackets comprises side rails configured to be received within the slots of a respective receptacle along the frame.

19. The method of orienting shots of claim 16, wherein: each of the first ballast and the second ballast comprises a support member at its distal end; each of the first and second bearing members defines (i) a tubular body comprising an inner diameter that receives a respective support member of the first ballast and the second ballast, and serving as a race, and (ii) a plurality of roller bearings along the race; and the method further comprises: placing the first bearing member onto the support member of the first ballast, and placing the second bearing member onto the support member of the second ballast.

20. The method of orienting shots of claim 16, wherein: the perforating gun assembly further comprises a first flange residing at a first end of the rail, and a second flange residing at a second end of the rail; the first flange comprises at least four spaced-apart through-openings; the second flange also comprises at least four spaced-apart through-openings; the proximal end of the first ballast comprises a pair of through-openings that may be aligned with a selected pair of the spaced-apart through-openings of the first flange; and the proximal end of the second ballast comprises a pair of through-openings that may be aligned with a selected pair of the spaced-apart through-openings of the second flange.

21. The method of orienting shots of claim 20, wherein connecting the proximal end of the first ballast to the first end of the rail, and connecting the proximal end of the second ballast to the second end of the rail, comprises: (i) placing first connectors through aligned through-openings between the proximal end of the first ballast and the first end of the rail; and (ii) placing second connectors through aligned through-openings between the proximal end of the second ballast and the second end of the rail.

22. The method of orienting shots of claim 16, wherein: a first electrically conductive contact pin resides within the bore of the first tandem sub; a second electrically conductive contact pin resides within the bore of the second tandem sub; a detonator resides within the gun barrel housing adjacent the rail; and the explosive charges may be placed within respective receptacles along the frame to fire at an orientation of any of 0 degrees, 90 degrees, 180 degrees, or 270 degrees within the horizontal portion of the wellbore.

23. The method of orienting shots of claim 16, further comprising: running the perforating gun assembly into the wellbore at an end of an electric wireline; pumping the perforating gun assembly into the horizontal portion of the wellbore; and allowing the weighted body of the first ballast and the weighted body of the second ballast to rotate into a downward position, thereby placing the explosive charges into a desired orientation relative to the wellbore.

24. A method of firing explosive charges in a hydrocarbon producing field, comprising: locating a parent wellbore in the hydrocarbon producing field; locating a child wellbore in the hydrocarbon

producing field; running a perforating gun assembly into the child wellbore, wherein the perforating gun assembly comprises: a gun barrel housing; a tandem sub placed at each of opposing ends of the gun barrel housing; and a rail system residing within the gun barrel housing, wherein the rail system comprises: a rail defining an elongated frame, one or more receptacles placed along the frame, an explosive charge residing within each of the respective one or more receptacles, and a ballast secured to each of opposing ends of the rail; positioning each explosive charge within its respective slot in a selected direction; securing the ballasts to the rail at a selected angle relative to the explosive charges; running the perforating gun assembly into the child wellbore at an end of an electric wireline; pumping the perforating gun assembly into a horizontal portion of the child wellbore such that the perforating gun assembly resides within a subsurface formation at a selected depth; and allowing the ballasts to rotate into a downward position.

25. The method of firing explosive charges of claim 24, wherein allowing the ballasts to rotate places each of the explosive charges into a position to fire charges in a horizontal orientation within the subsurface formation.

26. The method of firing explosive charges of claim 24, wherein allowing the ballasts to rotate places each of the explosive charges into a position to fire charges in a vertical orientation within the subsurface formation.

27. The method of firing explosive charges of claim 24, wherein: the charges may be received within a respective receptacle from either side of the frame; and the method further comprises selecting a direction for each of the explosive charges to be inserted into a respective receptacle along the frame.

28. The method of firing explosive charges of claim 27, further comprising: locating a direction of the parent wellbore within the field; and securing the ballasts to the respective opposing ends of the rail such that when the ballasts rotate within the horizontal portion of the child wellbore, the explosive charges will fire into the subsurface formation in a direction away from the parent wellbore.
